Model Predictive Control

Chapter 1c: System Theory Basics

Prof. Melanie Zeilinger

ETH Zurich

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Coauthors: Prof. Manfred Morari, University of Pennsylvania
Prof. Colin Jones, EPFL

Requirements for MPC

Need

1. A model of the system
2. A state estimator
3. Define the optimal control problem
4. Set up the optimization problem
5. Get the optimal control sequence (solve the optimization problem)
6. Verify that the closed-loop system performs as desired

Will repeat some of the basics on modeling and linear systems in this lecture.

Learning Objectives

- Describe dynamics with continuous-time state-space models
- Derive linearized model and understand limitations of linear description
- Discretize nonlinear and linear systems and contrast model properties
- Analyse stability, controllability, observability of linear systems
- Understand Lyapunov stability and prove stability of nonlinear systems
- Construct a Lyapunov function for stable linear systems

Outline

1. Models of Dynamic Systems

2. Analysis of LTI Discrete-Time Systems

3. Analysis of Nonlinear Discrete-Time Systems

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Models of Dynamic Systems

Goal: Introduce mathematical models to be used in Model Predictive Control (MPC) for describing the behavior of dynamic systems

- Model classification: state space/transfer function, linear/nonlinear, time-varying/time-invariant, continuous-time/discrete-time, deterministic/stochastic
- If not stated differently, we use deterministic models
- Models of physical systems derived from first principles are mainly: nonlinear, time-invariant, continuous-time, state space models (*)
- Target models for standard MPC are mainly: linear, time-invariant, discrete-time, state space models (†)
- Focus of this section is on how to 'transform' (*) to (†)

Abbreviations used in the following:
LTI = linear, time-invariant; DT = discrete time; CT = continuous time

Outline

1. Models of Dynamic Systems

   Nonlinear Time-Invariant Continuous-Time State Space Models

   LTI Continuous-Time State Space Models

   Time-Invariant, Discrete-Time, State Space Models

Nonlinear, Time-Invariant, Continuous-Time, State Space Models (1/3)

$$\dot{x} = g(x, u)$$
$$y = h(x, u)$$

$x \in \mathbb{R}^n$	state vector	$g(x, u) : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$	system dynamics
$u \in \mathbb{R}^m$	input vector	$h(x, u) : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^p$	output function
$y \in \mathbb{R}^p$	output vector		

- Very general class of models
- Higher order ODEs can be easily brought to this form (next slide)
- Analysis and control synthesis generally hard ($\to$ linearization to bring it to linear, time-invariant (LTI), continuous-time, state space form)

Nonlinear Time-Invariant Continuous-Time State Space Models (2/3)

Equivalence of one n-th order ODE and n 1-st order ODEs

$$x^{(n)} + g_n(x, \dot{x}, \ddot{x}, \ldots, x^{(n-1)}) = 0$$

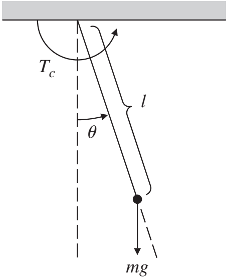
Define

$$x_{j+1} = x^{(j)}, \quad j = 0, \ldots, n-1$$

Transformed system

$$\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= x_3 \\
&\vdots \\
\dot{x}_{n-1} &= x_n \\
\dot{x}_n &= -g_n(x_1, x_2, \ldots, x_n)
\end{aligned}$$

Nonlinear Time-Invariant Continuous-Time State Space Models (3/3)



Example: Pendulum

Moment of inertia wrt. rotational axis	$m\,l^2$
Torque caused by external force	T_c
Torque caused by gravity	$m\,g\,l\sin(\theta)$

System equation

$$m\,l^2\,\ddot{\theta} = T_c - m\,g\,l\sin(\theta)$$

Using $x_1 := \theta, x_2 := \dot{\theta} = \dot{x}_1$ and $u := T_c/m\,l^2$ the system can be brought to standard form

$$\dot{x} = \begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{g}{l}\sin(x_1) + u\end{bmatrix} = g(x, u)$$

Output equation depends on the measurement configuration, i.e. if θ is measured then $y = h(x, u) = x_1$.

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LTI Continuous-Time State Space Models (1/7)

$$\dot{x} = A^c x + B^c u$$
$$y = Cx + Du$$

$x \in \mathbb{R}^n$	state vector	$A^c \in \mathbb{R}^{n \times n}$	system matrix
$u \in \mathbb{R}^m$	input vector	$B^c \in \mathbb{R}^{n \times m}$	input matrix
$y \in \mathbb{R}^p$	output vector	$C \in \mathbb{R}^{p \times n}$	output matrix
		$D \in \mathbb{R}^{p \times m}$	throughput matrix

- Vast theory exists for the analysis and control synthesis of linear systems
- Exact solution (next slide)

LTI Continuous-Time State Space Models (2/7)

Solution to linear ODEs

- Consider the ODE (written with explicit time dependence) $\dot{x}(t) = A^c x(t) + B^c u(t)$ with initial condition $x_0 := x(t_0)$, then its solution is given by

$$x(t) = e^{A^c(t-t_0)}x_0 + \int_{t_0}^{t} e^{A^c(t-\tau)} B^c u(\tau)\mathrm{d}\tau$$

  where $e^{A^c t} := \sum_{n=0}^{\infty} \frac{(A^c t)^n}{n!}$

LTI Continuous-Time State Space Models (3/7)

- **Problem:** Most physical systems are nonlinear but linear systems are much better understood
- Nonlinear systems can be well approximated by a linear system in a 'small' neighborhood around a point in state space
- **Idea:** Control keeps the system around some operating point
  $\rightarrow$ replace nonlinear by a linearized system around operating point

First order Taylor expansion of function $f(\cdot)$ around $\bar{x}$

$$f(x) \approx f(\bar{x}) + \left.\frac{\partial f}{\partial x^\top}\right|_{x=\bar{x}} (x - \bar{x}), \text{ with } \frac{\partial f}{\partial x^\top} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

LTI Continuous-Time State Space Models (4/7)

Linearization

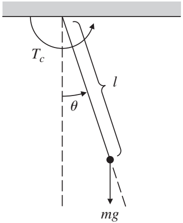
Stationary operating point x_s, u_s: $\dot{x}_s = g(x_s, u_s) = 0, y_s = h(x_s, u_s)$

$$
\begin{aligned}
\dot{x} &= \underbrace{g(x_s, u_s)}_{=0} + \underbrace{\left.\frac{\partial g}{\partial x^\top}\right|_{\substack{x=x_s\\u=u_s}}}_{=A^c} \underbrace{(x - x_s)}_{=\Delta x} + \underbrace{\left.\frac{\partial g}{\partial u^\top}\right|_{\substack{x=x_s\\u=u_s}}}_{=B^c} \underbrace{(u - u_s)}_{=\Delta u} \\
\Rightarrow \dot{x} - \underbrace{\dot{x}_s}_{=0} &= \Delta\dot{x} = A^c \Delta x + B^c \Delta u \\
y &= \underbrace{h(x_s, u_s)}_{y_s} + \underbrace{\left.\frac{\partial h}{\partial x^\top}\right|_{\substack{x=x_s\\u=u_s}}}_{=C} (x - x_s) + \underbrace{\left.\frac{\partial h}{\partial u^\top}\right|_{\substack{x=x_s\\u=u_s}}}_{=D} (u - u_s) \\
\Rightarrow \underbrace{\Delta y}_{y-y_s} &= C\Delta x + D\Delta u
\end{aligned}
$$

Subsequently, instead of Δx, Δu and Δy, x, u and y are used for brevity.

LTI Continuous-Time State Space Models (6/7)

Example: Linearization of pendulum equations



$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{g}{l}\sin(x_1) + u \end{bmatrix} = g(x, u)$$

$$y = x_1 = h(x, u)$$

Case 1: Want to keep the pendulum around $x_s = [\pi/4, 0]^\top \rightarrow u_s = \frac{g}{l}\sin(\pi/4)$

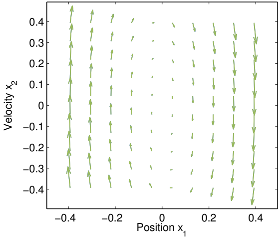
$$A^c = \left.\frac{\partial g}{\partial x^\top}\right|_{\substack{x=x_s \\ u=u_s}} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l}\cos(\pi/4) & 0 \end{bmatrix}, \quad B^c = \left.\frac{\partial g}{\partial u^\top}\right|_{\substack{x=x_s \\ u=u_s}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

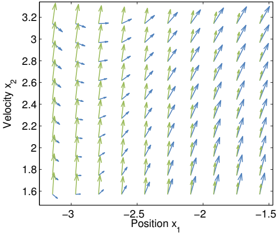
$$C = \left.\frac{\partial h}{\partial x^\top}\right|_{\substack{x=x_s \\ u=u_s}} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = \left.\frac{\partial h}{\partial u^\top}\right|_{\substack{x=x_s \\ u=u_s}} = 0$$

LTI Continuous-Time State Space Models (7/7)

Case 2: Want to keep the pendulum around $x_s = \begin{bmatrix} 0 & 0 \end{bmatrix}^\top$:

For feedback $u = -x_2$ linearization results in $\dot{x} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -1 \end{bmatrix} x$





Gradient of
nonlinear system: blue
linearized system: green

Linearization provides good approximation for small angles and velocities.

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    LTI Continuous-Time State Space Models

    Time-Invariant, Discrete-Time, State Space Models

Time-Invariant, Discrete-Time, State Space Models

- Discrete-time systems are described by difference equations

$$
\begin{aligned}
x(k+1) &= g(x(k), u(k)) \\
y(k) &= h(x(k), u(k))
\end{aligned}
$$

$x \in \mathbb{R}^n$	state vector	$g(x, u) : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$	system dynamics
$u \in \mathbb{R}^m$	input vector	$h(x, u) : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^p$	output function
$y \in \mathbb{R}^p$	output vector		

- Inputs and outputs of a discrete-time system are defined only at discrete time points, i.e. its inputs and outputs are sequences defined for $k \in \mathbb{Z}^+$
- Discrete time systems describe either
  1. Inherently discrete systems, eg. bank savings account balance at the k-th month

  $$x(k+1) = (1 + \alpha)x(k) + u(k)$$

  2. 'Transformed' continuous-time system

Time-Invariant, Discrete-Time, State Space Models

- Vast majority of controlled systems not inherently discrete-time systems
- Controllers almost always implemented using microprocessors
- Finite computation time must be considered in the control system design $\rightarrow$ **discretize** the continuous-time system
- Discretization is the procedure of obtaining an 'equivalent' discrete-time system from a continuous-time system
- The discrete-time model describes the state of the continuous-time system only at particular instances t_k, $k \in \mathbb{Z}^+$ in time, where $t_{k+1} = t_k + T_s$ and T_s is called the sampling time
- Usually $u(t) = u(t_k) \;\; \forall t \in [t_k, t_{k+1})$ is assumed (and implemented)

Euler Discretization of Nonlinear, Time-Invariant Models

- Given continuous time model

$$\dot{x}^c(t) = g^c(x^c(t), u^c(t))$$
$$y^c(t) = h^c(x^c(t), u^c(t))$$

- Approximate $\dot{x}^c(t) \approx \frac{x^c(t+T_s) - x^c(t)}{T_s}$
- T_s is the **sampling time**
- Notation: $x(k) := x^c(t_0 + kT_s)$, $u(k) := u^c(t_0 + kT_s)$
- Then the DT model is

$$x(k+1) = x(k) + T_s g^c(x(k), u(k)) = g(x(k), u(k))$$
$$y(k) = h^c(x(k), u(k)) = h(x(k), u(k))$$

- Under regularity assumptions, if T_s is small and CT and DT have 'same' initial conditions and inputs, then outputs of CT and DT systems 'will be close'

Example: Euler Discretization of the Nonlinear Pendulum Equations

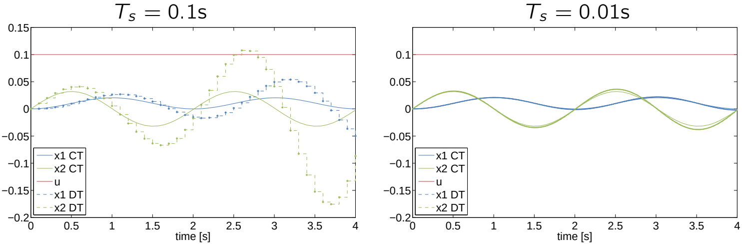
Using $g/l = 10[s^{-2}]$ the nonlinear pendulum equations are given by

$$\dot{x} = \begin{bmatrix} x_2 \\ -10\sin(x_1) + u \end{bmatrix}$$

Discretizing the continuous time system using Euler we get the following discrete time system

$$x(k+1) = x(k) + T_s \begin{bmatrix} x_2(k) \\ -10\sin(x_1(k)) + u(k) \end{bmatrix}$$

Results for two different sampling times:



Euler Discretization of Linear, Time-Invariant Models

- Given CT model

$$\begin{aligned} \dot{x}(t) &= A^c x(t) + B^c u(t) \\ y(t) &= C^c x(t) + D^c u(t) \end{aligned}$$

- The DT model obtained with Euler discretization is

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

with $A = I + T_s A^c$, $B = T_s B^c$, $C = C^c$, $D = D^c$.

- There are a variety of discretization approaches (matlab: help c2d)

Exact Discretization of Linear Time-Invariant Models

- Recall the solution of the ODE $x(t) = e^{A^c(t-t_0)}x_0 + \int_{t_0}^{t} e^{A^c(t-\tau)}Bu(\tau)\mathrm{d}\tau$
- Choose $t_0 = t_k$ (hence $x_0 = x(t_k)$), $t = t_{k+1}$ and use $t_{k+1} - t_k = T_s$ and $u(t) = u(t_k) \ \ \forall t \in [t_k, t_{k+1})$

$$
\begin{aligned}
x(t_{k+1}) &= e^{A^c T_s} x(t_k) + \int_{t_k}^{t_{k+1}} e^{A^c(t_{k+1}-\tau)} B^c \mathrm{d}\tau \, u(t_k) \\
&= \underbrace{e^{A^c T_s}}_{\triangleq A} x(t_k) + \underbrace{\int_0^{T_s} e^{A^c(T_s-\tau')} B^c \mathrm{d}\tau'}_{\triangleq B} \, u(t_k) \\
&= Ax(t_k) + Bu(t_k)
\end{aligned}
$$

- We found the **exact** discrete-time model predicting the state of the continuous-time system at time t_{k+1} given $x(t_k)$, $k \in \mathbb{Z}_+$ under the assumption of a constant $u(t)$ during a sampling interval
- $B = (A^c)^{-1}(A - I)B^c$, if A^c invertible

Exact Discretization of Linear Time-Invariant Models

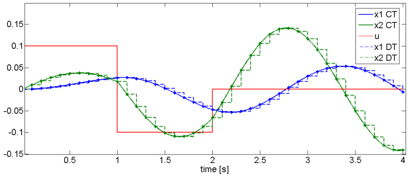
Example: Discretization of the linearized pendulum equations

Using $g/l = 10[s^{-2}]$ the pendulum equations linearized about $x_s = \begin{bmatrix}\pi/4 & 0\end{bmatrix}^\top$ are given by

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -10/\sqrt{2} & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

Discretizing the continuous-time system using the definitions of A and B, and $T_s = 0.1$ s, we get the following discrete-time system

$$x(k+1) = \begin{bmatrix} 0.965 & 0.099 \\ -0.699 & 0.965 \end{bmatrix} x(k) + \begin{bmatrix} 0.005 \\ 0.100 \end{bmatrix} u(k)$$



Solution of Linear Discrete Time Systems

Given:
- The discrete time linear system $x(k+1) = Ax(k) + Bu(k)$
- An initial state $x(k)$ at discrete time k
- An input sequence $\{u(k), \ldots, u(k+N-1)\}$

Find the solution of the discrete time system at time $k+N$:

$$
\begin{aligned}
x(k+1) &= Ax(k) + Bu(k) \\
x(k+2) &= Ax(k+1) + Bu(k+1) = A(Ax(k) + Bu(k)) + Bu(k+1) \\
&= A^2x(k) + ABu(k) + Bu(k+1) \\
x(k+3) &= Ax(k+2) + Bu(k+2) \\
&= A^3x(k) + A^2Bu(k) + ABu(k+1) + Bu(k+2) \\
&\vdots \\
x(k+N) &= A^Nx(k) + A^{N-1}Bu(k) + \cdots + ABu(k+N-2) + Bu(k+N-1)
\end{aligned}
$$

and therefore $x(k+N) = A^Nx(k) + \sum_{i=0}^{N-1} A^iBu(k+N-1-i)$

$\Rightarrow$ Linear function of the initial state and the input sequence

In Summary: We Work with Discrete Time Models

We will use:

- Nonlinear Discrete Time Models

$$\begin{aligned} x(k+1) &= g(x(k), u(k)) \\ y(k) &= h(x(k), u(k)) \end{aligned}$$

- or Linear Time-Invariant Discrete Time Models

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

Discretization

We call **discretization** the procedure of obtaining an "equivalent" DT model from a CT one.

Models of Dynamic Systems: Recap

Continuous time systems:

- Continuous time state space models provide framework for representing general class of models
- Analysis and control synthesis is generally hard
- Linearization of nonlinear system to facilitate controller design and analysis
- Linearization only valid in a small neighborhood of the linearization point
- Continuous time system requires system integration to predict model behavior

Discrete time systems:

- Discrete time models most suited for MPC
- Solution of discrete time model is a linear function of the initial state and the input sequence
- Zero-order-hold provides exact discretization for linear systems
- For general nonlinear systems only approximate discretization methods exist, such as Euler discretization, quality depends on sampling time

Outline

1. Models of Dynamic Systems
2. Analysis of LTI Discrete-Time Systems
3. Analysis of Nonlinear Discrete-Time Systems

Outline

2. Analysis of LTI Discrete-Time Systems

    Coordinate Transformations

    Stability

    Observability & Controllability

Coordinate Transformations (1/2)

- Consider again the system

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

- Input-output behavior, i.e. the sequence $\{y(k)\}_{k=0,1,2\ldots}$ entirely defined by $x(0)$ and $\{u(k)\}_{k=0,1,2\ldots}$
- Infinitely many choices of the state that yield the same input-output behavior
- Certain choices facilitate system analysis

Coordinate Transformations (2/2)

- Consider the linear transformation $\tilde{x}(k) = Tx(k)$ with $\mathbf{det}(T) \neq 0$ (invertible)

$$\begin{aligned} T^{-1}\tilde{x}(k+1) &= AT^{-1}\tilde{x}(k) + Bu(k) \\ y(k) &= CT^{-1}\tilde{x}(k) + Du(k) \end{aligned}$$

or

$$\begin{aligned} \tilde{x}(k+1) &= \underbrace{TAT^{-1}}_{\tilde{A}}\tilde{x}(k) + \underbrace{TB}_{\tilde{B}}u(k) \\ y(k) &= \underbrace{CT^{-1}}_{\tilde{C}}\tilde{x}(k) + \underbrace{D}_{\tilde{D}}u(k) \end{aligned}$$

- Note: $u(k)$ and $y(k)$ are unchanged

Outline

2. Analysis of LTI Discrete-Time Systems

    Coordinate Transformations

    Stability

    Observability & Controllability

Stability of Linear Systems (1/3)

Theorem: Asymptotic Stability of Linear Systems

The LTI system

$$x(k+1) = Ax(k)$$

is globally asymptotically stable

$$\lim_{k \to \infty} x(k) = 0, \forall x(0) \in \mathbb{R}^n$$

if and only if $|\lambda_j| < 1$, $\forall j = 1, \cdots, n$ where λ_j are the eigenvalues of A.[1]

[1] for cont. time LTI systems $\dot{x} = Ax$, the condition is $Re(\lambda_j) < 0$

Stability of Linear Systems (2/3)

"Proof" of asymptotic stability condition

- Assume that A has n linearly independent eigenvectors $e_1, \cdots, e_n$ then the coordinate transformation $\tilde{x}(k) = [e_1, \cdots, e_n]^{-1} x(k) = Tx(k)$ transforms an LTI discrete-time system to

$$\tilde{x}(k+1) = TAT^{-1}\tilde{x}(k) = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \vdots \\ \vdots & 0 & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix} \tilde{x}(k) = \Lambda \tilde{x}(k)$$

- The state $\tilde{x}(k)$ can be explicitly formulated as a function of $\tilde{x}(0) = Tx(0)$

$$\tilde{x}(k) = \Lambda^k \tilde{x}(0) = \begin{bmatrix} \lambda_1^k & 0 & \cdots & 0 \\ 0 & \lambda_2^k & 0 & \vdots \\ \vdots & 0 & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n^k \end{bmatrix} \tilde{x}(0)$$

Stability of Linear Systems (3/3)

"Proof" of asymptotic stability condition

- We thus have that

$$\begin{aligned} \tilde{x}(k) = \Lambda^k \tilde{x}(0) \quad &\Rightarrow \quad |\tilde{x}(k)| = |\Lambda^k \tilde{x}(0)| \quad \text{(component-wise)} \\ &\Rightarrow \quad |\tilde{x}(k)| = |\Lambda^k| \cdot |\tilde{x}(0)| \\ &\Rightarrow \quad |\tilde{x}_j(k)| = |\lambda_j^k| \cdot |\tilde{x}_j(0)| = |\lambda_j|^k \cdot |\tilde{x}_j(0)| \end{aligned}$$

- If any $|\lambda_j| \geq 1$ then $\lim_{k\to\infty} \tilde{x}(k) \neq 0$ for $\tilde{x}(0) \neq 0$. On the other hand if $|\lambda_j| < 1 \; \forall j \in 1, \cdots, n$ then $\lim_{k\to\infty} \tilde{x}(k) = 0$ and we have asymptotic stability
- If the system does not have n linearly independent eigenvectors it can not be brought into diagonal form and Jordan matrices have to be used for the proof but the assertions still hold

Outline

2. Analysis of LTI Discrete-Time Systems

   Coordinate Transformations

   Stability

   Observability & Controllability

Controllability (1/3)

Controllability

A system $x(k+1) = Ax(k) + Bu(k)$ is **controllable**[a] if for any pair of states $x(0), x^*$ there exists a finite time N and a control sequence $\{u(0), \cdots, u(N-1)\}$ such that $x(N) = x^*$,i.e.

$$x^* = x(N) = A^N x(0) + \begin{bmatrix} B & AB & \cdots & A^{N-1}B \end{bmatrix} \begin{bmatrix} u(N-1) \\ u(N-2) \\ \vdots \\ u(0) \end{bmatrix}$$

[a] Often referred to as “reachable" for discrete time systems.

It follows from the **Cayley-Hamilton** theorem that A^k can be expressed as linear combinations of $A^j, j \in 0, 1, \cdots, n-1$ for $k \geq n$. Hence for all $N \geq n$

$$\text{range}\begin{bmatrix} B \ AB \ \cdots A^{N-1}B \end{bmatrix} = \text{range}\begin{bmatrix} B \ AB \ \cdots A^{n-1}B \end{bmatrix}$$

Controllability (2/3)

- If the system cannot be controlled to x^* in n steps, then it cannot in an arbitrary number of steps
- Define the **controllability matrix** $\mathcal{C} = \begin{bmatrix} B \ AB \ \cdots A^{n-1}B \end{bmatrix}$
- The system is controllable if

$$\mathcal{C} \begin{bmatrix} u(n-1) \\ u(n-2) \\ \vdots \\ u(0) \end{bmatrix} = x^* - A^n x(0)$$

  has a solution for all right-hand sides (RHS)
- From linear algebra: solution exists for all RHS iff n columns of $\mathcal{C}$ are linearly independent

Proposition:
Necessary and and sufficient condition for controllability is $\text{rank}(\mathcal{C}) = n$

Controllability (3/3)

Remarks

- Another related concept is stabilizability
- A system is called **stabilizable** if there exists an input sequence that returns the state to the origin asymptotically, starting from an arbitrary initial state
- A system is stabilizable iff all of its uncontrollable modes are stable
- Stabilizability can be checked using the following condition

$$\text{if } \operatorname{rank}\left([\lambda_j I - A \mid B]\right) = n \quad \forall \lambda_j \in \Lambda_A^+ \;\Rightarrow\; (A, B) \text{ is stabilizable}$$

  where Λ_A^+ is the set of all eigenvalues of A lying on or outside the unit circle.
- Controllability implies stabilizability

Observability (1/3)

Consider the following system with zero input

$$\begin{aligned} x(k+1) &= Ax(k) \\ y(k) &= Cx(k) \end{aligned}$$

Observability
A system is said to be **observable** if there exists a finite N such that for every $x(0)$ the measurements $y(0), y(1), \cdots y(N-1)$ uniquely distinguish the initial state $x(0)$

Observability (2/3)

- Question of uniqueness of the linear equations

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1) \end{bmatrix} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{N-1} \end{bmatrix} x(0)$$

- As previously we can replace N by n wlog. (**Cayley-Hamilton**)
- Define $\mathcal{O} = \left[C^\top \ (CA)^\top \ \cdots \ (CA^{n-1})^\top \right]^\top$
- From linear algebra: solution is unique iff the n columns of $\mathcal{O}$ are linearly independent

Proposition:

Necessary and sufficient condition for observability of system (A, C) is $\text{rank}(\mathcal{O}) = n$

Observability (3/3)

Remarks

- Another related concept is detectability
- A system is called **detectable** if it possible to construct from the measurement sequence a sequence of state estimates that converges to the true state asymptotically, starting from an arbitrary initial estimate
- A system is detectable iff all of its unobservable modes are stable
- Detectability can be checked using the following condition

  $$\text{if } \operatorname{rank}\left(\left[A^\top - \lambda_j I \mid C^\top\right]\right) = n \quad \forall \lambda_j \in \Lambda_A^+ \;\Rightarrow\; (A, C) \text{ is detectable}$$

  where Λ_A^+ is the set of all eigenvalues of A lying on or outside the unit circle.
- Observability implies detectability

Outline

1. Models of Dynamic Systems

2. Analysis of LTI Discrete-Time Systems

3. Analysis of Nonlinear Discrete-Time Systems

Outline

3. Analysis of Nonlinear Discrete-Time Systems

   Stability of Nonlinear Systems

   Lyapunov Stability of LTI Discrete-Time Systems

   Example

Stability of Nonlinear Systems (1/6)

- We consider first the stability of a nonlinear, time-invariant, discrete-time system
$$x(k+1) = g(x(k)) \tag{1}$$
with an *equilibrium point* at $\bar{x}$, i.e. $g(\bar{x}) = \bar{x}$.
- For nonlinear systems there are many definitions of stability.
- Informally, we define a system to be stable in the sense of Lyapunov, if it stays in any arbitrarily small neighborhood of the equilibrium when it is disturbed slightly.
- In the following we always mean "stability" in the sense of Lyapunov.
- Note that system (1) encompasses any open- or closed-loop autonomous system.
- We will then derive simpler stability conditions for the specific case of LTI systems.
- Note that stability is a property of an equilibrium point of a system.

Stability of Nonlinear Systems (2/6)

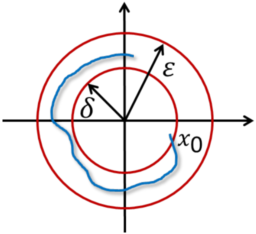
Lyapunov stability

Formally, the equilibrium point $\bar{x}$ of a system (1) is

- **Lyapunov stable** if for every $\epsilon > 0$ there exists a $\delta(\epsilon) > 0$ such that

$$\|x(0) - \bar{x}\| < \delta(\epsilon) \rightarrow \|x(k) - \bar{x}\| < \epsilon, \forall k \geq 0$$

- **unstable** otherwise.



Global asymptotic stability

An equilibrium point $\bar{x}$ of system (1) is **globally asymptotically stable if it is Lyapunov stable and** attractive**, i.e.,**

$$\lim_{k \to \infty} \|x(k) - \bar{x}\| = 0, \ \forall\, x(0) \in \mathbb{R}^n.$$

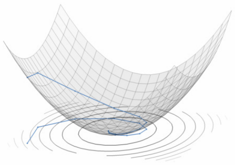
Stability of Nonlinear Systems (3/6)

- We can show stability by constructing a **Lyapunov function**
- Idea: A mechanical system is asymptotically stable, if the total mechanical energy is decreasing over time (friction losses). A Lyapunov function is a system theoretic generalization of energy.

Definition: Global Lyapunov function

Consider the equilibrium point $\bar{x} = 0$ of system (1). A function $V : \mathbb{R}^n \to \mathbb{R}$, continuous at the origin, finite for every $x \in \mathbb{R}^n$, and such that

$$
\begin{aligned}
\|x\| \to \infty \Rightarrow V(x) \to \infty \\
V(0) = 0 \text{ and } V(x) > 0,\ \forall x \in \mathbb{R}^n \setminus \{0\} \\
V(g(x)) - V(x) \leq -\alpha(x)\ \forall x \in \mathbb{R}^n
\end{aligned}
$$



where $\alpha : \mathbb{R}^n \to \mathbb{R}$ is continuous positive definite,

is called a **Lyapunov function**.

Stability of Nonlinear Systems (4/6)

Lyapunov theorem

Theorem: Global Lyapunov stability (global asymptotic stability)
If a system (1) admits a Lyapunov function $V(x)$, then $x = 0$ is **globally asymptotically stable**.

Note: If $V(x)$ satisfies the condition $V(g(x)) - V(x) \leq -\alpha(x)\ \forall x \in \mathbb{R}^n$ only with $\alpha(x)$ positive semidefinite, then $x = 0$ is **globally Lyapunov stable**.

Stability of Nonlinear Systems (5/6)

Theorem: Lyapunov indirect method

Let $A = \frac{\partial g}{\partial x^\top}\Big|_{x=0}$ be the linearization matrix for system (1) around $\bar{x} = 0$. Then the origin is locally asymptotically stable if $|\lambda_i| < 1$ for all the eigenvalues of A. Instead, if there exists at least one eigenvalue such that $|\lambda_i| > 1$, then the origin is unstable.

- If there exists at least one eigenvalue $|\lambda_i| = 1$, then we cannot say anything about stability $\rightarrow$ construct a Lyapunov function.
- The theorem is valid also for continuous-time systems. The eigenvalue conditions will change accordingly: $|\lambda_i| < 1 \rightarrow Re(\lambda_i) < 0$ and $|\lambda_i| > 1 \rightarrow Re(\lambda_i) > 0$

Stability of Nonlinear Systems (6/6)

Remarks

- Note that the Lyapunov theorems only provide sufficient conditions
- Lyapunov theory is a powerful concept for proving stability of a control system, but for general nonlinear systems it is usually difficult to find a Lyapunov function
- Lyapunov functions can sometimes be derived from physical considerations
- One common approach:
  - Decide on **form** of Lyapunov function (e.g., quadratic)
  - Search for parameter values e.g. via optimization so that the required properties hold
- Will see: MPC allows for a stability guarantee for constrained systems: Lyapunov function is provided in the form of the optimal cost function (under certain assumptions on the problem setup).
- For linear systems there exist constructive theoretical results on the existence of a quadratic Lyapunov function

Outline

3. Analysis of Nonlinear Discrete-Time Systems

Stability of Nonlinear Systems

Lyapunov Stability of LTI Discrete-Time Systems

Example

Global Lyapunov Stability of Linear Systems (1/3)

- Consider the linear system

$$x(k+1) = Ax(k) \tag{2}$$

- Take $V(x) = x^\top Px$ with $P > 0$ (positive definite) as a candidate Lyapunov function. It satisfies $V(0) = 0$, $V(x) > 0$ and $\|x\| \to \infty \Rightarrow V(x) \to \infty$.
- Check 'energy decrease' condition

$$\begin{aligned} V(Ax(k)) - V(x(k)) &= x^\top(k)A^\top PAx(k) - x^\top(k)Px(k) \\ &= x^\top(k)(A^\top PA - P)x(k) \leq -\alpha(x(k)) \end{aligned}$$

- We can choose $\alpha(x(k)) = x^\top(k)Qx(k)$, $Q > 0$. Hence, the condition can be satisfied if a $P > 0$ can be found that solves the **discrete-time Lyapunov equation**

$$A^\top PA - P = -Q,\ \ Q > 0. \tag{3}$$

Global Lyapunov Stability of Linear Systems (2/3)

Theorem: Existence of solution to the DT Lyapunov equation

The discrete-time Lyapunov equation (3) has a unique solution $P > 0$ if and only if A has all eigenvalues inside the unit circle, i.e. if and only if the system $x(k+1) = Ax(k)$ is stable.

- Therefore, for LTI systems global asymptotic Lyapunov stability is not only sufficient but also necessary, and it agrees with the notion of stability based on eigenvalue location.
- Note that stability is always "global" for linear systems.

Global Lyapunov Stability of Linear Systems (3/3)

Property of P

- The matrix P can also be used to determine the infinite horizon cost-to-go for an asymptotically stable autonomous system $x(k+1) = Ax(k)$ with a quadratic cost function determined by Q.
- More precisely, defining $\Psi(x(0))$ as

$$\Psi(x(0)) = \sum_{k=0}^{\infty} x(k)^\top Q x(k) = \sum_{k=0}^{\infty} x(0)^\top (A^k)^\top Q A^k x(0) \tag{4}$$

  we have that

$$\Psi(x(0)) = x(0)^\top P x(0). \tag{5}$$

"Proof"

- Define $H_k := (A^k)^\top Q A^k$ and $P := \sum_{k=0}^{\infty} H_k$ (limit of the sum exists because the system is assumed asymptotically stable).
- We have that $A^\top H_k A = (A^{k+1})^\top Q A^{k+1} = H_{k+1}$.
- Thus
  $A^\top P A = \sum_{k=0}^{\infty} A^\top H_k A = \sum_{k=0}^{\infty} H_{k+1} = \sum_{k=1}^{\infty} H_k = P - H_0 = P - Q$.

Outline

3. Analysis of Nonlinear Discrete-Time Systems

   Stability of Nonlinear Systems

   Lyapunov Stability of LTI Discrete-Time Systems

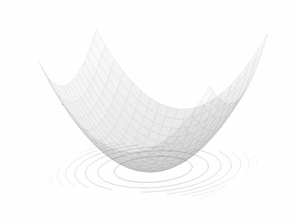
   Example

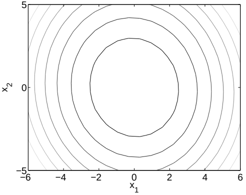
Example: Lyapunov Stability for Linear Systems

- Consider the discrete time system $x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} u(k)$ with $u(k) = Kx(k) = -\begin{bmatrix} 0.1160 & 0.5385 \end{bmatrix} x(k)$
- We choose $Q = I$ and solve the discrete time Lyapunov equation $(A+BK)^\top P(A+BK) - P = -Q$:

$$P = \begin{bmatrix} 3.3921 & 2.7757 \\ 2.7757 & 7.7136 \end{bmatrix}$$

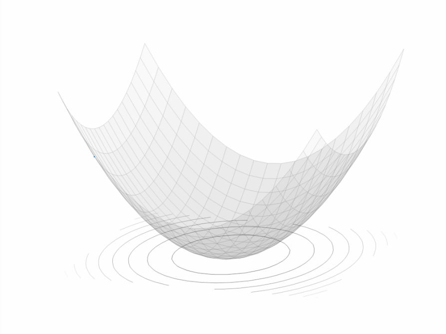
- Since $V(x) = x^\top Px$ is a quadratic function, all level sets are ellipsoids with shape matrix P

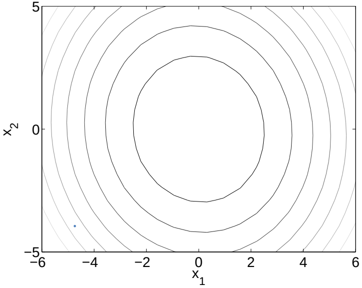




Example: Lyapunov Stability for Linear Systems

- Consider the discrete time system $x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} u(k)$ with $u(k) = Kx(k) = -\begin{bmatrix} 0.1160 & 0.5385 \end{bmatrix} x(k)$
- Closed-loop simulation:





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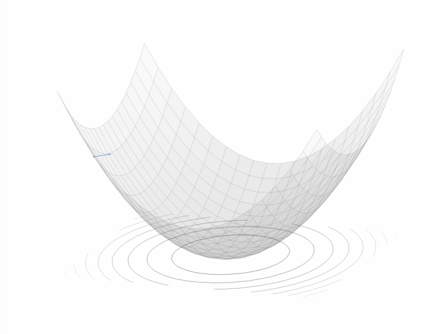

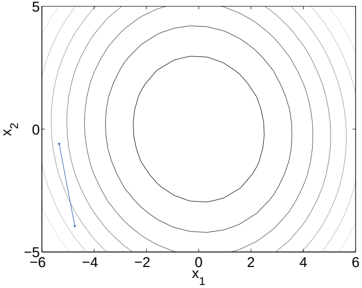

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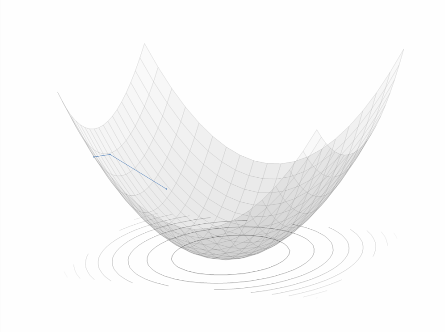

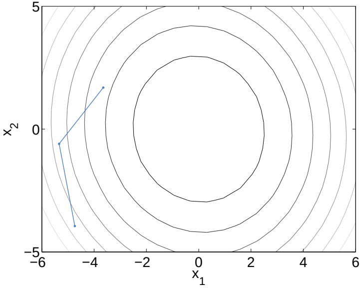

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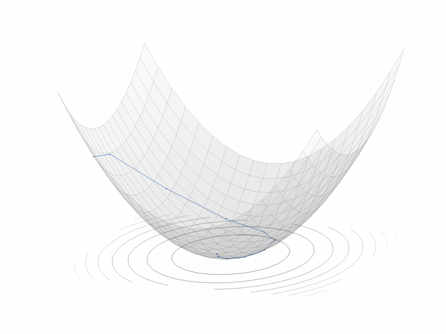

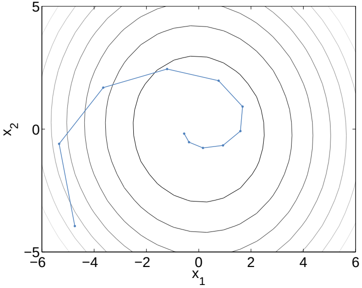

Analysis of Discrete Time Systems: Recap

- Lyapunov stability provides framework to analyze stability of general nonlinear systems
- Asymptotic stability captures convergence in the limit and is commonly used in MPC
- For LTI systems, asymptotic stability can be shown by eigenvalues of the dynamic matrix
- Lyapunov functions provide means to prove stability for nonlinear systems
- Key is how to find a Lyapunov function
- For linear systems there exists a quadratic Lyapunov function if and only if the system is stable
- MPC allows to prove stability by providing a Lyapunov function for the closed-loop system
- Controllability/Observability are necessary requirements for design of a controller/estimator (can be relaxed into stabilizability and detectability)